04. Nmap Scan Types and Packet Behaviour

1. What Is Nmap?

Definition: Nmap (Network Mapper) is a free network discovery and security auditing tool used to identify hosts, open ports, services and operating-system clues.

Key points:

- Discovers active hosts.
- Identifies open, closed and filtered ports.
- Detects services and versions.
- Can estimate the operating system.
- Should only be used on authorized systems.

2. Seven Important Nmap commands

Option	Name and Purpose	Typical Network Behaviour
-sn	Host discovery without a port scan	May use ARP, ICMP or TCP probes, depending on network and privileges
-sS	TCP SYN scan	Sends SYN probes; open ports often reply SYN-ACK, closed ports often reply RST
-sT	TCP connect scan	Uses the operating system's connect() call to complete TCP connections
-sU	UDP scan	Sends UDP probes; replies or ICMP port-unreachable messages help classify ports
-sV	Service/version detection	Sends additional probes to identify service names and versions
-O	OS detection	Analyses network responses to estimate the target operating system
-A	Aggressive detection bundle	Enables OS detection, version detection, default scripts and traceroute

3. Understanding SYN Scans

In a SYN scan, Nmap sends TCP SYN packets to selected ports.

Typical responses:

- SYN-ACK: the port is likely open.
- RST: the port is likely closed.
- No response or certain ICMP errors: the port may be filtered.

A SYN scan usually does not complete a normal TCP handshake for open ports. However, it still generates network traffic and may be recorded by firewalls, IDS tools or endpoint/network monitoring systems.

4. Connect Scans vs SYN Scans

Feature	**SYN Scan**	**Connect Scan**
Option	-sS	-sT
Connection	Usually does not complete the TCP handshake	Completes the TCP connection
Privileges	Usually requires raw-packet privileges	Uses the operating system's connection API
Logs	May be recorded by network security tools	May also appear in service or connection logs
Use	Port discovery	Port discovery when raw-packet access is unavailable

5. Safe Lab Examples

Use these commands only against your own Ubuntu and Windows VMs.

Host discovery:
nmap -sn 192.168.56.0/24

SYN scan:
sudo nmap -sS -p 22,80,443 192.168.56.20

Connect scan:
nmap -sT -p 22,80,443 192.168.56.20

UDP scan:
sudo nmap -sU -p 53,123 192.168.56.20

Service detection:
nmap -sV 192.168.56.20

OS detection:
sudo nmap -O 192.168.56.20

Aggressive detection:
sudo nmap -A 192.168.56.20

Replace the example IP with the target VM's actual lab IP address.

6. Saving Nmap Results

Normal text output:
nmap -sS -oN syn_scan.txt 192.168.56.20

XML output:
nmap -sS -oX syn_scan.xml 192.168.56.20

Both formats:
nmap -sS -oA syn_scan 192.168.56.20

The -oA option creates output in normal, XML and grepable formats.

7. What Should an Analyst Look For?

- One source contacting many ports on a single host.
- One source scanning many hosts.
- Repeated SYN packets with few completed connections.
- Unexpected scans against administrative services.
- Scans originating from an unusual device.

A scan can be legitimate administration or troubleshooting. Confirm the source, timing and authorization before classifying it as malicious.

8. Conclusion

Nmap helps discover hosts, ports, services and operating-system clues. Understanding how different scan types generate packets helps analysts recognize scanning patterns in Wireshark and network security logs.